1. Probability & Set Theory

Exercise:
Roll a fair six-sided die ($\Omega = \{1, 2, 3, 4, 5, 6\}$). Find $P(\text{roll} < 4)$.

Solution:

- Outcomes less than 4 are: $\{1, 2, 3\}$.
- Total outcomes: 6.
- Thus,
$$P(\text{roll} < 4) = \frac{3}{6} = \frac{1}{2} = 0.5.$$

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2. Operations with Events

Exercise:
If $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$, find:

1. $P(A^c)$
2. $P(A \cup B)$
3. $P((A \cup B)^c)$

Solution:

1. $P(A^c) = 1 - P(A) = 1 - 0.4 = 0.6$.
2. Using the inclusion-exclusion principle,
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.3 - 0.1 = 0.6.$$
3. $P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.6 = 0.4$.

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3. Conditional Probability & Independence

Exercise:
Two fair coin tosses: Show that “First toss = Heads” and “Second toss = Heads” are independent.

Solution:

- Let A = “First toss = Heads” and B = “Second toss = Heads”.
- For a fair coin, $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{2}$.

- Since the tosses are independent, the joint probability is

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

- Also, $P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

- Since $P(A \cap B) = P(A)P(B)$, the events are independent.

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5. Discrete vs. Continuous RV Examples

Exercise:
A fair coin is tossed 5 times. Let X be the number of heads. Find $P(X = 2)$ and $E[X]$.

Solution:

- X follows a Binomial distribution with $n = 5$ and $p = 0.5$.

- The probability mass function is:

$$P(X = k) = \binom{5}{k}(0.5)^k(0.5)^{5-k} = \binom{5}{k}(0.5)^5.$$

- For $k = 2$:

$$P(X = 2) = \binom{5}{2}(0.5)^5 = \frac{10}{32} = \frac{5}{16} \approx 0.3125.$$

- The expected value for a Binomial distribution is:

$$E[X] = np = 5 \times 0.5 = 2.5.$$

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6. Normal Distribution & Applications

Exercise:
If $X \sim N(100, 16)$, find $P(90 \leq X \leq 110)$ using Z-scores or tables.

Solution:

- Mean $\mu = 100$ and variance $\sigma^2 = 16$ so $\sigma = 4$.

- For $X = 90$:

$$z = \frac{90 - 100}{4} = -\frac{10}{4} = -2.5.$$

- For $X = 110$:

$$z = \frac{110 - 100}{4} = \frac{10}{4} = 2.5.$$

- Now,

$$P(90 \leq X \leq 110) = P(-2.5 \leq Z \leq 2.5).$$

- Using the standard normal table,

$$P(Z \leq 2.5) \approx 0.9938 \quad \text{and} \quad P(Z \leq -2.5) \approx 0.0062.$$

- Therefore,

$$P(-2.5 \leq Z \leq 2.5) = 0.9938 - 0.0062 = 0.9876.$$

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7. Basic Statistical Inference

Exercise:
You sample 100 people's heights. Which is the parameter, and which is the statistic?

Solution:

- The **parameter** is a numerical characteristic of the entire population (e.g., the true mean height μ or true standard deviation σ).

- The **statistic** is a numerical summary computed from the sample (e.g., the sample mean $\bar{X}$ or sample standard deviation s).

- Thus, in this case, μ is the parameter and $\bar{X}$ is the statistic.

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8. Point Estimation & Confidence Intervals

Exercise:
A sample of size $n = 25$ from a normal population has $\bar{X} = 50$ and $s = 4$. Construct a 95% confidence interval for μ.

Solution:

- Since the population is normal and the sample size is small, use the t-distribution with $n - 1 = 24$ degrees of freedom.

- The 95% confidence interval is given by:

$$\bar{X} \pm t_{\alpha/2,\, n-1} \cdot \frac{s}{\sqrt{n}}.$$

- From the t-table, for 24 d.o.f. at the 95% confidence level, $t_{0.025,24} \approx 2.064$.

- Compute the margin of error:

$$\text{Margin of Error} = 2.064 \times \frac{4}{\sqrt{25}} = 2.064 \times \frac{4}{5} = 2.064 \times 0.8 \approx 1.6512.$$

- Thus, the 95% confidence interval for μ is:

$$50 \pm 1.65,$$

  which is approximately:

$$(48.35,\ 51.65).$$